NAME_____ EXAMINATION NO _____

ZOMBA CATHOLIC CLUSTER MOCK

2019 MALAWI SCHOOL CERTIFICATE OF EDUCATION

AGRICULTURE

PAPER I

(100 MARKS)

Date **Subject Number:** M012/I

Time Allowed 2hrs

Instructions

1. This paper contains **9 pages**. Please check
2. This paper contains two sections **A and B**. answer all questions from all sections.
3. In the table provided on this page, tick against the question number you have answered
4. Write your School, name and class on all pages
5. Hand in your paper to the invigilator when time is called to stop writing

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			

Turn over

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SECTION A (70 MARKS)

1. a) Give two form of soil degradation.

_____

_____ (2 marks)

b. Explain any two ways in which poor farming practices encourage

_____

_____

_____

_____ (4 marks)

c. Explain any one way in which each of the following can modify soil PH

(i) Use of organic manure

_____

_____ (2 marks)

(ii) Use of sulphate of Ammonia

_____

_____ (2 marks)

2. **Table 1** shows an experiment to find the effect of adding different fertilizers to the yield of potatoes and wheat crops. Use it to answer the question that follow

Plot	**Fertilizer Added in kg/ha**			**Crop yield/ha**	
	N	**P**	**K**	**Wheat**	**Potatoes**
A	0	0	0	1.7	9.5
B	96	0	0	3.7	8.3
C	0	77	107	2.0	16.7
D	96	77	107	6.6	38.6

a. (i) Which plot was control?

_____ (1 mark)

(ii) Give a reason to the answer 2a (i) above

_____ (1 mark)

b. What is the effect of adding only **N** fertilizer on the yield of potatoes

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_____ (1 mark)

c. Use the data in the table to describe the effect on the yield of wheat of adding **P** and **K**

(i) Without **N**

_____ (1 mark)

(ii) With **N**

_____ (1 mark)

d. Explain any one way in which this experiment is important in crop production.

_____

_____ (2 marks)

3. a) Explain any one way in which each of the following irrigation problems can be solved following irrigation problems can be solved

a. Water logging

_____

_____ (2 marks)

b.Salt accumulation

_____

_____ (2 marks)

4. The demand for beef in a certain market changed from 7500kg to 5500kg when the price changed from K110 to K120 per kg

a.Calculate the price of Elasticity of Demand

(4 marks)

b.State the degree of elasticity of demand for beef at this market

_____ (1 mark)

c. What I implication would the degree of price Elasticity of demand for beef at this market have on beef providers

_____

_____ (2 marks)

5. a) Outline any two examples of Vegetative planting materials

_____

_____ (2 marks)

c. state any two advantages and disadvantages of using vegetative planting materials

(i) Advantages

_____

_____ (2 marks)

(ii) Disadvantages

_____

_____ (2 marks)

6. A maize plant is showing the following symptoms of nutrient deficiency in a field:
Scotched of leaf margins, small dots on the leaves and weak stalks.

a. Identify the nutrient that the maize plant is lacking

_____ (1 mark)

b. State any two ways of correcting the deficiency symptoms

_____

_____ (2 marks)

c. Give three ways how the nutrient is depleted from the soil.

_____

_____

_____ (3 marks)

7. Explain any one way in which the following improved farming technologies would ensure food security for the growing population

a. use of herbicides

_____

_____ (2 marks)

b. use of irrigation

_____

_____ (2 marks)

8. a) List any two husbandry practices for mushroom production

_____

_____ (2 marks)

b. Explain any two importance of mushroom production in Malawi

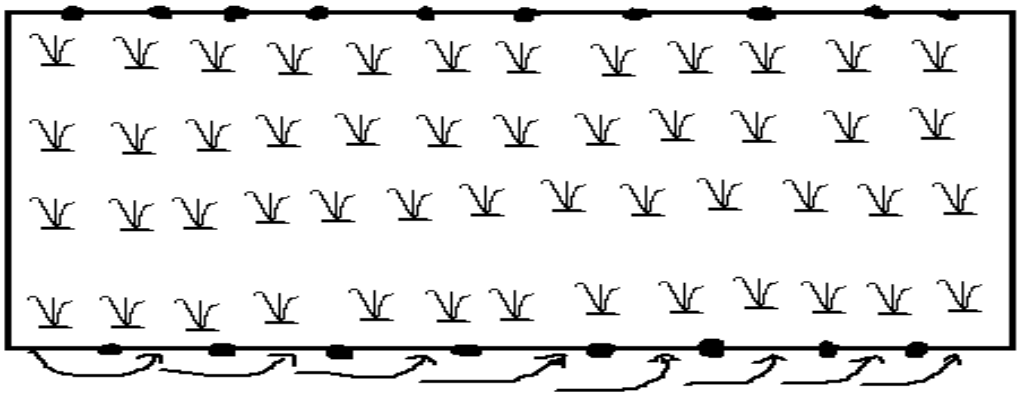
_____

_____

_____

_____ (4 marks)

9. **Fig 1** below shows a diagram of one of the systems of grazing animals. Use it to answer the questions that follows:



a. Name the system of grazing animals

_____ (1 mark)

b. Explain the two advantages of the above system of grazing animals

_____

_____

_____

_____ (4 marks)

c. Explain why the system of grazing animals is not commonly used in Malawi

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_____

_____ (2 marks)

d.Give any three examples of legume pasture

_____

_____

_____ (3 marks)

10. Table 2 belows shows information on green maize and groundnuts. Use it to answer the question that follow;

Item	Groundnuts	Green maize
G/nuts: 100 bag@ 3,000 per bag Green maize: 10,000 cobs@ K50 per cob	300,000	500,000
Total Revenue	300,000	500,000
Total variable costs	25,000	35,000
Total fixed costs	15,000	20,000

a. Calculate each of the following

(i) Breakeven quantity for the green maize

_____

_____

_____ (3 marks)

(ii) Breakeven price for the groundnuts

_____

_____

_____ (3 marks)

b. State any two ways in which calculation of breakeven budget is important in farm business management

_____

_____ (2 marks)

c. Describe the importance of Farm Records

_____

_____ (2 marks)

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SECTION B (30 MARKS)

Answer all questions in this section

11. Explain any **five** stages in Silage making

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_____

_____(10 marks)

12. Explain any **five** importance of trading at community level

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_____

_____(10 marks)

13. Explain any **five** ways how Intensive farming systems support the growing population

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_____

_____(10 marks)

END OF QUESTION PAPER